



Machine Learning 1

Lecture 3 - Linear Regression - Basis functions
- Maximum Likelihood - Stochastic Gradient
Descent - Underfitting and Overfitting -
Regularized Least Squares

Erik Bekkers

(Bishop 3.1)



Intended Learning Outcome

- ▶ Contrast **MAP** estimation with **full Bayesian inference**, and recognize a **conjugate prior**
- ▶ Write a **linear model with basis functions**, and explain why it is still called linear
- ▶ Derive **least squares as maximum likelihood** under Gaussian noise
- ▶ Solve it with the normal equations, or with **SGD** when that is too expensive
- ▶ **Diagnose overfitting**, and derive **ridge regression as MAP estimation**
- ▶ Write down a predictive distribution, and say what integrating the weights out buys over a point estimate



Machine Learning 1

Lecture 2.3 - Maximum Likelihood

Erik Bekkers

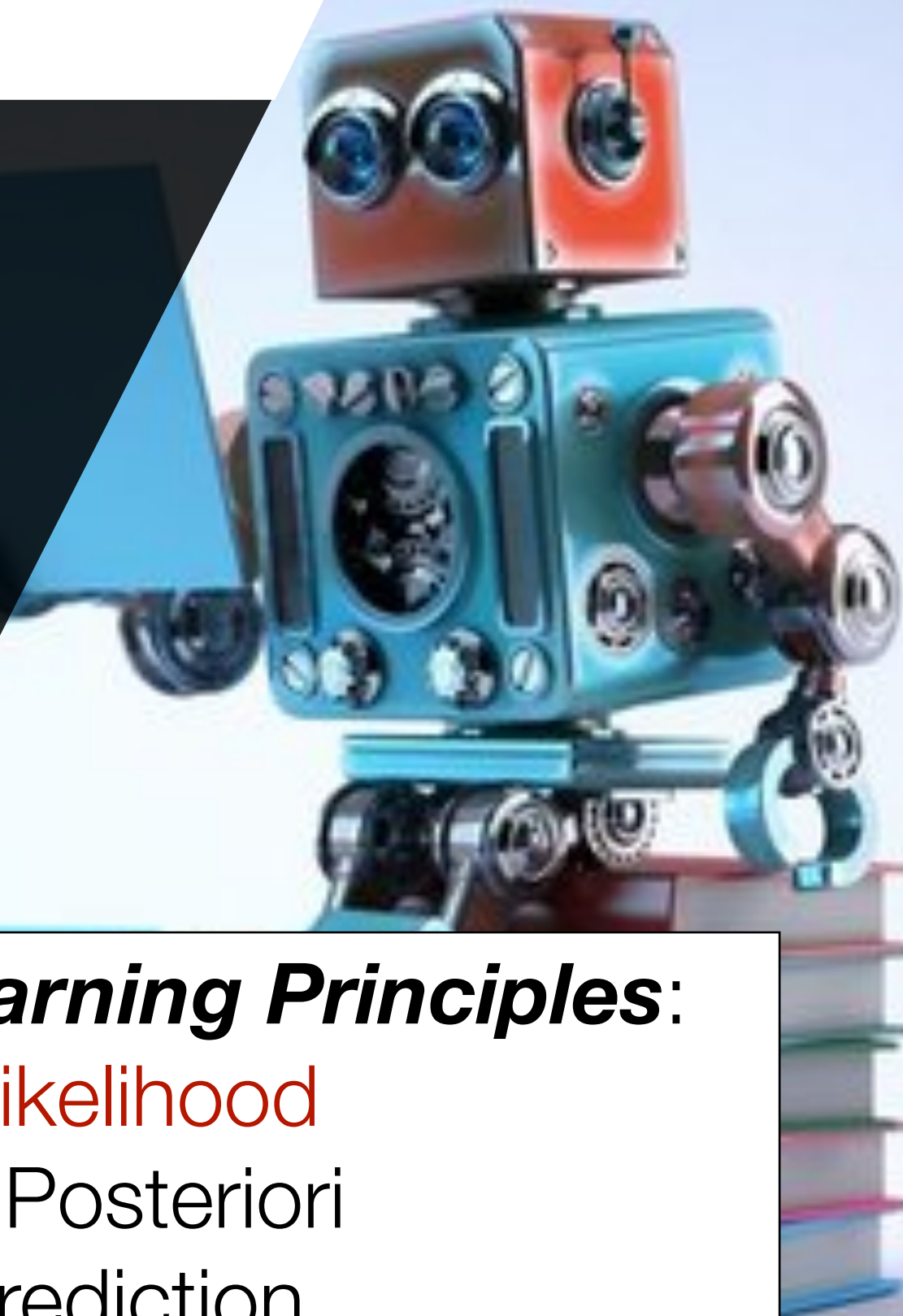
(Bishop 1.2.3 - 1.2.5)

Three Statistical Learning Principles:

Maximum Likelihood

Maximum A Posteriori

Bayesian Prediction



Maximum Likelihood Principle

- ▶ **Maximum Likelihood Optimisation**: Consider $D = \{x_1, x_2, \dots, x_N\} \sim p(D | \mathbf{w})$, assuming it came from some parametric distribution $p(D | \mathbf{w})$. Then solve:

$$\mathbf{w}_{ML} = \underset{\mathbf{w}}{\operatorname{argmax}} p(D | \mathbf{w}) \quad (\text{find parameters } \mathbf{w} \text{ that maximise the likelihood } p(D | \mathbf{w}))$$

- ▶ Assume samples in D to be **independent, identically distributed** (i.i.d), meaning each $x_i \sim p(x_i | \mathbf{w})$:

$$\begin{aligned} p(D | \mathbf{w}) &= p(x_1, x_2, \dots, x_N | \mathbf{w}) \\ &= p(x_1 | \mathbf{w}) p(x_2 | \mathbf{w}) \dots p(x_N | \mathbf{w}) \\ &= \prod_{i=1}^N p(x_i | \mathbf{w}) \end{aligned}$$

- ▶ Define equivalent problem of minimising **negative log likelihood (NLL)**:

$$\mathbf{w}_{ML} = \underset{\mathbf{w}}{\operatorname{argmin}} -\log p(D | \mathbf{w}) \quad \xRightarrow{i.i.d.} \quad \mathbf{w}_{ML} = \underset{\mathbf{w}}{\operatorname{argmin}} -\sum_{i=1}^N \log p(x_i | \mathbf{w})$$

Maximum Likelihood Principle

<https://ideal-machine-intelligence.eu/teaching/ml1/explainers/maximum-likelihood.html>

Maximum Likelihood Principle

Analytic solutions for Gaussians, i.e., the following can be solved in closed form

$$\mathbf{w}_{ML} = \underset{\mathbf{w}}{\operatorname{argmin}} - \sum_{i=1}^N \log p(x_i | \mathbf{w})$$

The solution being:

μ_{ML} is Sample mean:

$$\mu_{ML} = \frac{1}{N} \sum_{i=1}^N x_i$$

σ_{ML}^2 is sample variance:

$$\sigma_{ML}^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu_{ML})^2$$

Is the estimator unbiased?

$$\mathbb{E}_{D \sim p(D|\mu, \sigma)}[\mu_{ML}] =$$

$$\mathbb{E}_{D \sim p(D|\mu, \sigma)}[\sigma_{ML}^2] =$$



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Lecture 2.4 - Maximum Likelihood: An Example

Erik Bekkers

(Bishop 1.2.3 - 1.2.5)



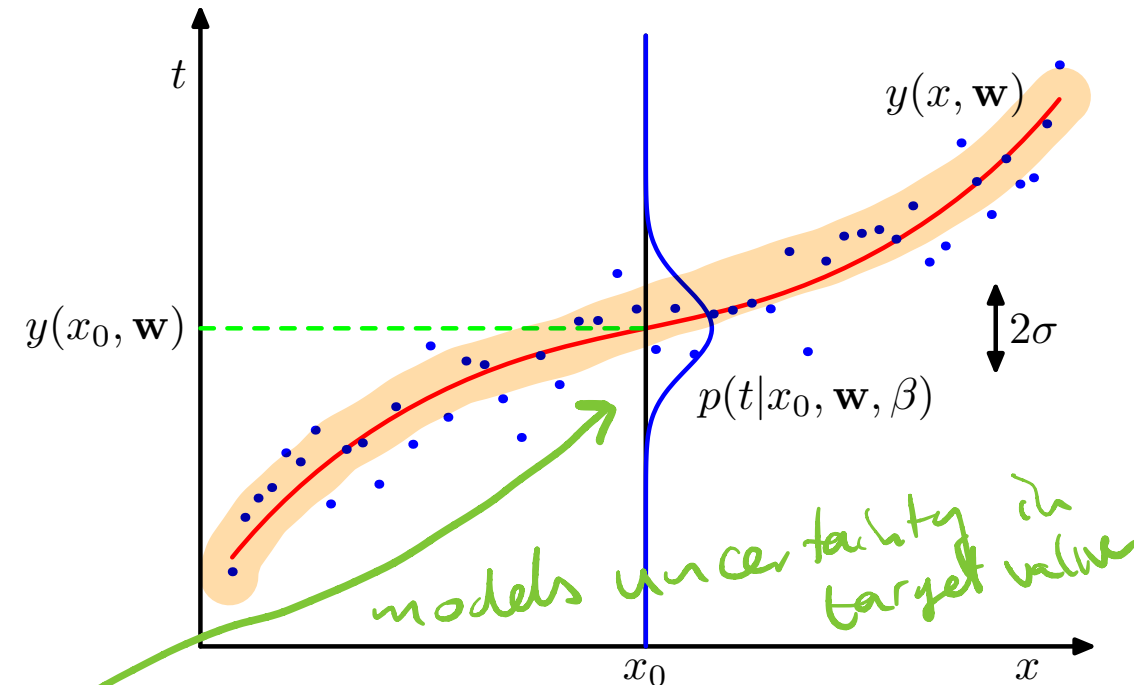
Curve Fitting: Maximum Likelihood Estimates

- ▶ Data $D = \{(x_1, t_1), \dots, (x_N, t_N)\} = \{\mathbf{x}, \mathbf{t}\}$
- ▶ Assume data is generated through underlying model

$$t = y(x, \mathbf{w}) + \sigma \epsilon$$

with Gaussian noise $\epsilon \sim \mathcal{N}(\mathbf{0}, 1)$

- ▶ Define **precision** β such that $\beta^{-1} = \sigma^2$



- ▶ Under these model assumptions we have **target distribution**:

$$p(t|x, \mathbf{w}, \beta) = \mathcal{N}(t | \underbrace{y(x, \mathbf{w})}_{\mu(x)}, \underbrace{\beta^{-1}}_{\sigma^2})$$

mean per position

- ▶ Log likelihood:

$$\begin{aligned} \log p(\mathbf{t} | \mathbf{x}, \mathbf{w}, \beta) &= \log \prod_i^N \mathcal{N}(t_i | y(x_i, \mathbf{w}), \beta^{-1}) \\ &= \frac{N}{2} \log \beta - \frac{N}{2} \log 2\pi - \frac{\beta}{2} \sum_{i=1}^N (t_i - y(x_i, \mathbf{w}))^2 \end{aligned}$$

Curve Fitting: Maximum Likelihood Estimates

- ML: minimize $E(\mathbf{x}, \mathbf{t}, \mathbf{w}, \beta) = -\log p(\mathbf{t} | \mathbf{x}, \mathbf{w}, \beta)$ w.r.t. $\mathbf{w}$ and β

$$E(\mathbf{x}, \mathbf{t}, \mathbf{w}, \beta) = \underbrace{\frac{\beta}{2} \sum_{i=1}^N (y(x_i, \mathbf{w}) - t_i)^2}_{\text{does not depend on } \beta} - \frac{N}{2} \log \beta + \frac{N}{2} \log 2\pi$$

- Maximum likelihood solution (least squares fit!):

$$\mathbf{w}_{ML} = \operatorname{argmin}_{\mathbf{w}} \frac{\beta}{2} \sum_{i=1}^N (y(x_i, \mathbf{w}) - t_i)^2$$

$$\frac{1}{\beta_{ML}} = \frac{1}{N} \sum_{i=1}^N (y(x_i, \mathbf{w}_{ML}) - t_i)^2$$

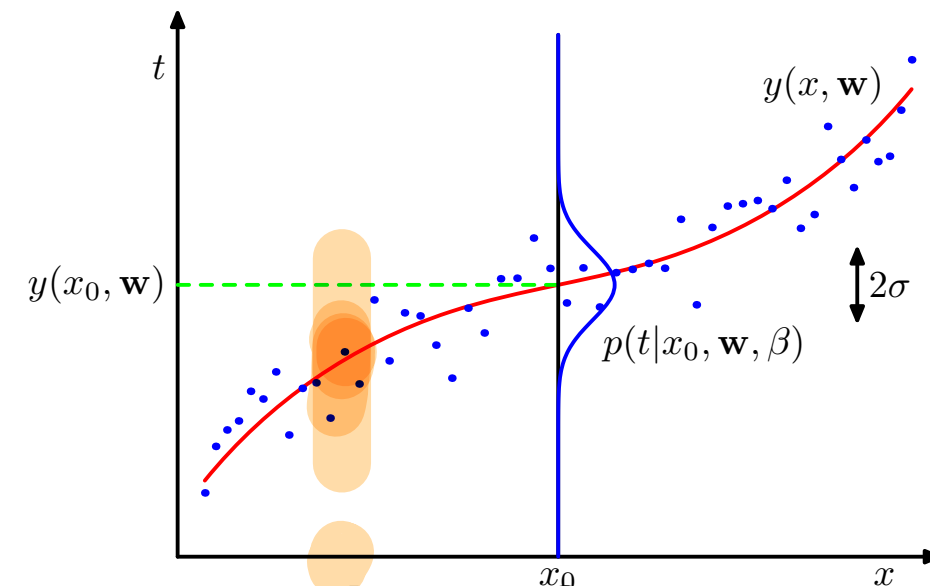


Figure: Gaussian conditional distribution (Bishop 1.16)

- Predictive distribution:**

best point estimate → $p(\underline{t'} | x', \mathbf{w}_{ML}, \beta_{ML}) = \mathcal{N}(t' | y(x', \mathbf{w}_{ML}), \beta_{ML}^{-1})$

→ $\mathbb{E}[t' | x', \mathbf{w}_{ML}, \beta_{ML}] = y(x', \mathbf{w}_{ML})$



Machine Learning 1

Lecture 2.5 - Maximum A Posteriori

Erik Bekkers

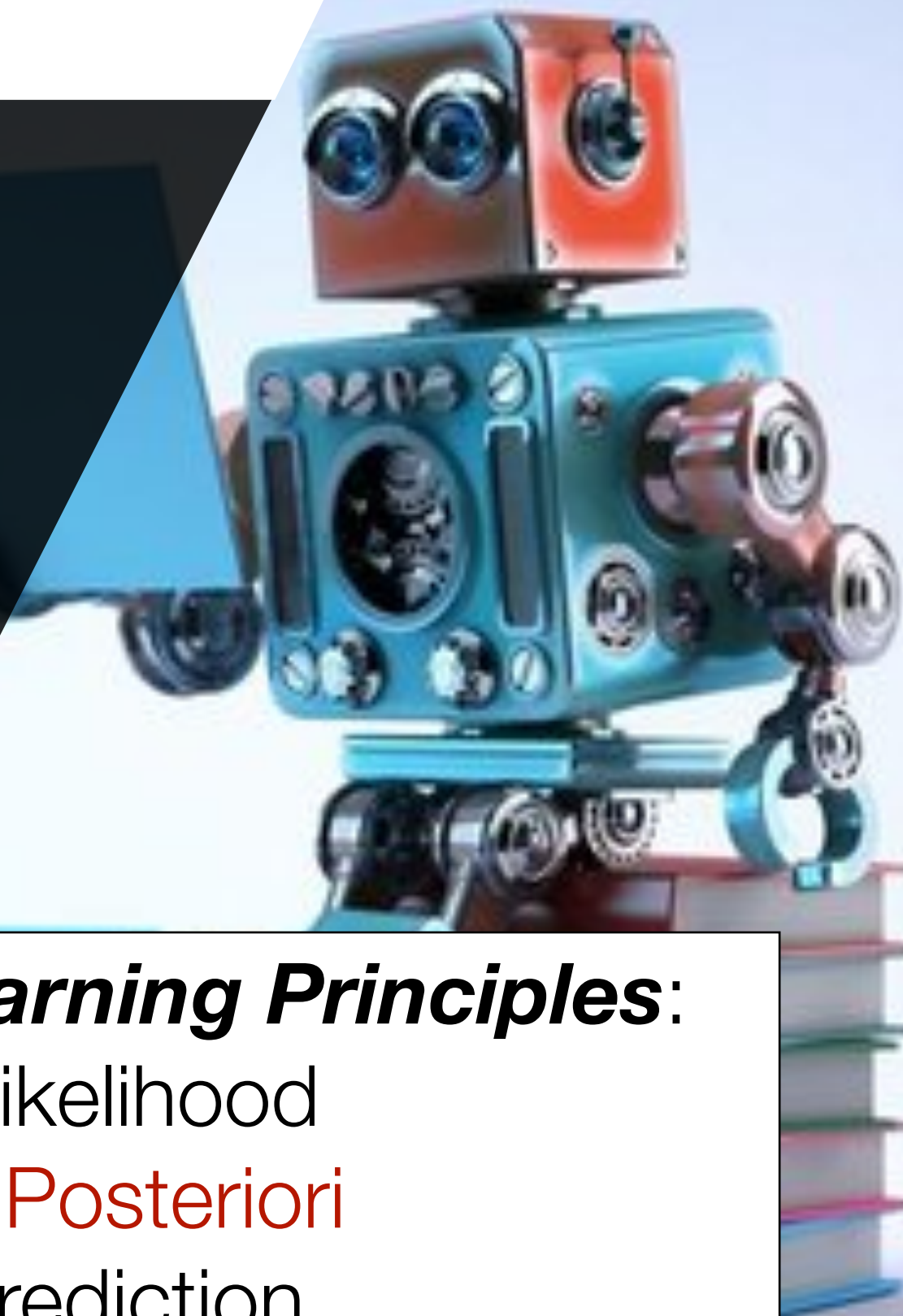
(Bishop 1.2.5 - 1.2.6)

Three Statistical Learning Principles:

Maximum Likelihood

Maximum A Posteriori

Bayesian Prediction



Maximum A Posteriori Estimates

- ▶ Dataset $D = (x_1, x_2, \dots, x_N)$ of N independent observations.
- ▶ **ML** estimate: choose $\mathbf{w}$ such that data likelihood is maximized:

$$\mathbf{w}_{ML} = \underset{\mathbf{w}}{\operatorname{argmax}} p(D | \mathbf{w})$$

- ▶ **MAP** estimate: choose most probable $\mathbf{w}$ given the data.

$$\mathbf{w}_{MAP} = \underset{\mathbf{w}}{\operatorname{argmax}} p(\mathbf{w} | D)$$

posterior probability

The posterior

- ▶ Dataset $D = (x_1, x_2, \dots, x_N)$ of N independent observations.
- ▶ Model the data distribution with some $p(D | \mathbf{w})$ parametrized by $\mathbf{w}$
- ▶ **MAP** estimate: choose **most probable** $\mathbf{w}$ given the data.

$$\mathbf{w}_{MAP} = \underset{\mathbf{w}}{\operatorname{argmax}} p(\mathbf{w} | D)$$

- ▶ With **posterior** $p(\mathbf{w} | D)$ obtained via Bayes' rule

$$p(\mathbf{w} | D) = \frac{p(D | \mathbf{w}) p(\mathbf{w})}{p(D)}$$

Maximum A Posteriori Estimates

- Dataset $D = (x_1, x_2, \dots, x_N)$ of N independent observations.
- Model the data distribution with some $p(D | \mathbf{w})$ parametrized by $\mathbf{w}$
- **MAP** estimate: choose **most probable $\mathbf{w}$** given the data.

$$\mathbf{w}_{MAP} = \underset{\mathbf{w}}{\operatorname{argmax}} p(\mathbf{w} | D)$$

Maximum A Posteriori Estimates

- ▶ Dataset $D = (x_1, x_2, \dots, x_N)$ of N independent observations.
- ▶ Model the data distribution with some $p(D | \mathbf{w})$ parametrized by $\mathbf{w}$
- ▶ **MAP** estimate: choose **most probable** $\mathbf{w}$ given the data.

$$\mathbf{w}_{MAP} = \operatorname{argmax}_{\mathbf{w}} p(\mathbf{w} | D)$$

$$\mathbf{w}_{MAP} = \operatorname{argmax}_{\mathbf{w}} \frac{p(D | \mathbf{w}) p(\mathbf{w})}{p(D)}$$

does not depend
on $\mathbf{w}$

Maximum A Posteriori Estimates

- ▶ Dataset $D = (x_1, x_2, \dots, x_N)$ of N independent observations.
- ▶ Model the data distribution with some $p(D | \mathbf{w})$ parametrized by $\mathbf{w}$
- ▶ **MAP** estimate: choose **most probable $\mathbf{w}$** given the data.

$$\mathbf{w}_{MAP} = \operatorname{argmax}_{\mathbf{w}} p(\mathbf{w} | D)$$

$$\mathbf{w}_{MAP} = \operatorname{argmax}_{\mathbf{w}} \frac{p(D | \mathbf{w}) p(\mathbf{w})}{p(D)}$$


$$\mathbf{w}_{MAP} = \operatorname{argmax}_{\mathbf{w}} p(D | \mathbf{w}) p(\mathbf{w})$$

Maximum A Posteriori Estimates

- ▶ Dataset $D = (x_1, x_2, \dots, x_N)$ of N independent observations.
- ▶ Model the data distribution with some $p(D | \mathbf{w})$ parametrized by $\mathbf{w}$
- ▶ **MAP** estimate: choose **most probable $\mathbf{w}$** given the data.

$$\mathbf{w}_{MAP} = \operatorname{argmax}_{\mathbf{w}} p(\mathbf{w} | D)$$

$$\mathbf{w}_{MAP} = \operatorname{argmax}_{\mathbf{w}} \frac{p(D | \mathbf{w}) p(\mathbf{w})}{p(D)}$$

$$\mathbf{w}_{MAP} = \operatorname{argmax}_{\mathbf{w}} p(D | \mathbf{w}) p(\mathbf{w})$$

$$\mathbf{w}_{MAP} = \operatorname{argmin}_{\mathbf{w}} - \log p(D | \mathbf{w}) - \log p(\mathbf{w})$$

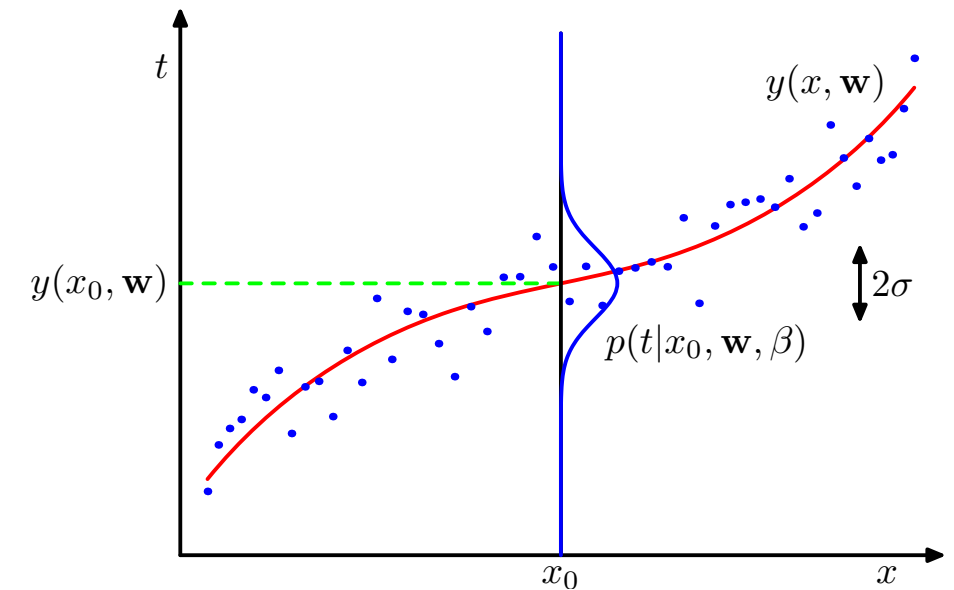
log brick

neg. log likelihood *log prior*

MAP - Example Gaussian distribution

- **Likelihood**/Data model:

$$\begin{aligned} p(t | x, \mathbf{w}, \beta) &= \mathcal{N}(t | y(x, \mathbf{w}), \beta^{-1}) \\ &= \sqrt{\frac{\beta}{2\pi}} \exp \left[-\frac{\beta}{2} (t - y(x, \mathbf{w}))^2 \right] \end{aligned}$$

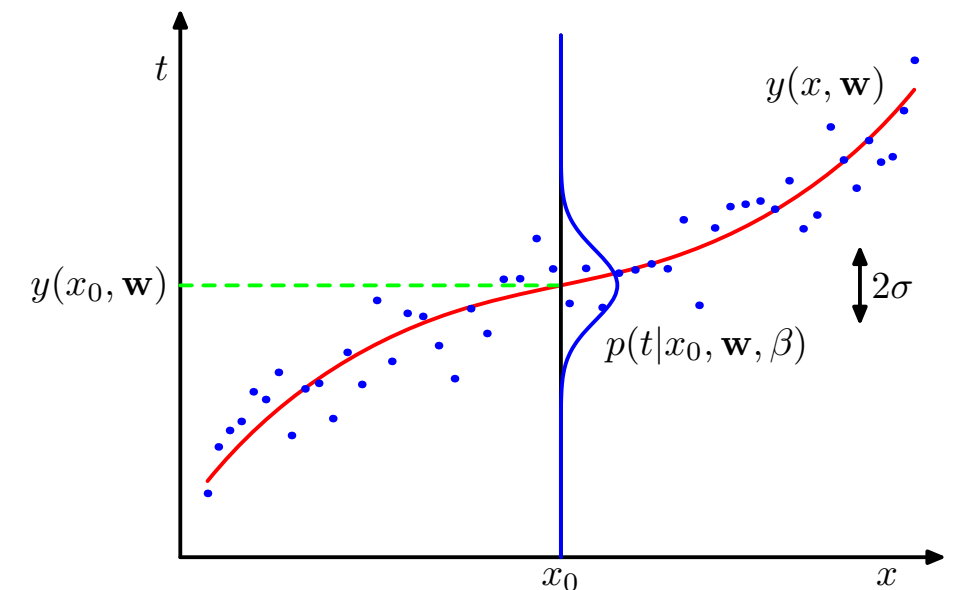


MAP - Example Gaussian distribution

- **Likelihood**/Data model:

$$p(t | x, \mathbf{w}, \beta) = \mathcal{N}(t | y(x, \mathbf{w}), \beta^{-1})$$

$$= \sqrt{\frac{\beta}{2\pi}} \exp \left[-\frac{\beta}{2} (t - y(x, \mathbf{w}))^2 \right]$$



- **Prior** (assumption: $\mathbf{w}$ takes on small values with certainty α):

$$p(\mathbf{w} | \alpha) = \prod_{i=1}^M \mathcal{N}(w_i | 0, \alpha^{-1})$$

$$= \left(\frac{\alpha}{2\pi} \right)^{M/2} \prod_{i=1}^M e^{-\frac{\alpha}{2} w_i w_i}$$

$$= \left(\frac{\alpha}{2\pi} \right)^{M/2} e^{-\frac{\alpha}{2} \mathbf{w}^T \mathbf{w}}$$

$$w_i \sim \mathcal{N}(0, \alpha^{-1})$$

$$e^{-\frac{\alpha}{2} \sum_{i=1}^M w_i w_i}$$

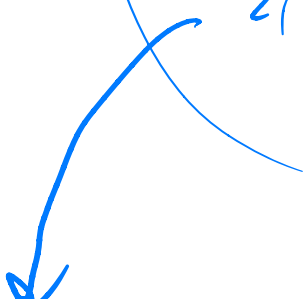
$$e^{-\mathbf{w}^T \Sigma^{-1} \mathbf{w}}$$

$$\Sigma = \alpha^{-1} \text{Id}$$

covariance

multi-variate

Gaussian $\mathcal{N}(\mathbf{0}, \alpha^{-1} \text{Id})$

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} & \Sigma_{13} \\ \Sigma_{21} & \Sigma_{22} & \Sigma_{23} \\ & & \ddots \end{pmatrix}$$


covariance $[w_1, w_2] = 0$

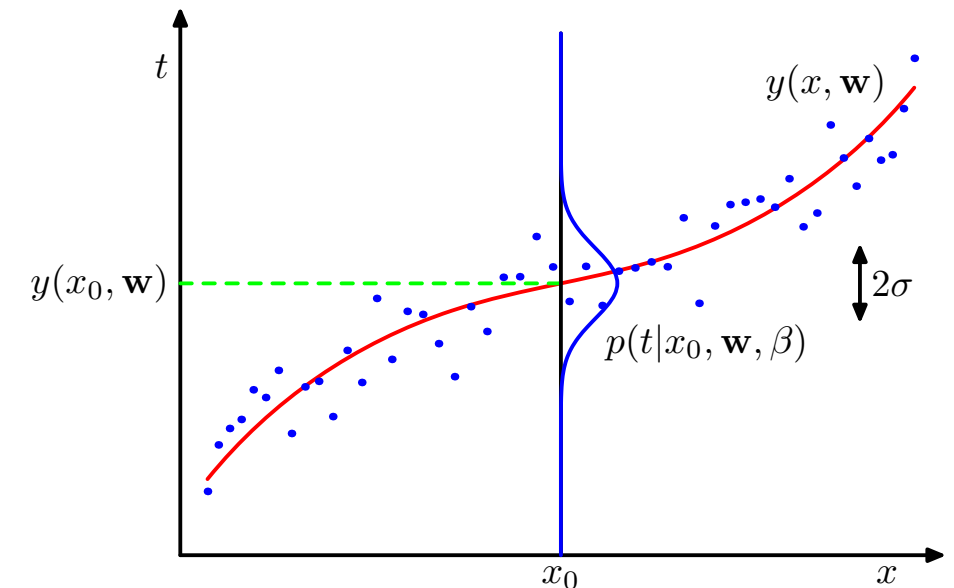
$$\Sigma = \begin{pmatrix} \Sigma_{11} & 0 & 0 \\ 0 & \Sigma_{22} & 0 \\ 0 & 0 & \Sigma_{33} \end{pmatrix}$$

$$= \begin{pmatrix} \alpha^{-1} & 0 & 0 \\ 0 & \alpha^{-1} & 0 \\ 0 & 0 & \alpha^{-1} \end{pmatrix} = \alpha^{-1} I_d$$

MAP - Example Gaussian distribution

- ▶ **Likelihood**/Data model:

$$\begin{aligned} p(t | x, \mathbf{w}, \beta) &= \mathcal{N}(t | y(x, \mathbf{w}), \beta^{-1}) \\ &= \sqrt{\frac{\beta}{2\pi}} \exp \left[-\frac{\beta}{2} (t - y(x, \mathbf{w}))^2 \right] \end{aligned}$$



- ▶ **Prior** (assumption: $\mathbf{w}$ takes on small values with certainty α):

$$\begin{aligned} p(\mathbf{w} | \alpha) &= \prod_{i=1}^M \mathcal{N}(w_i | 0, \alpha^{-1}) \\ &= \left(\frac{\alpha}{2\pi} \right)^{M/2} \prod_{i=1}^M e^{-\frac{\alpha}{2} w_i w_i} \\ &= \left(\frac{\alpha}{2\pi} \right)^{M/2} e^{-\frac{\alpha}{2} \mathbf{w}^T \mathbf{w}} \end{aligned}$$

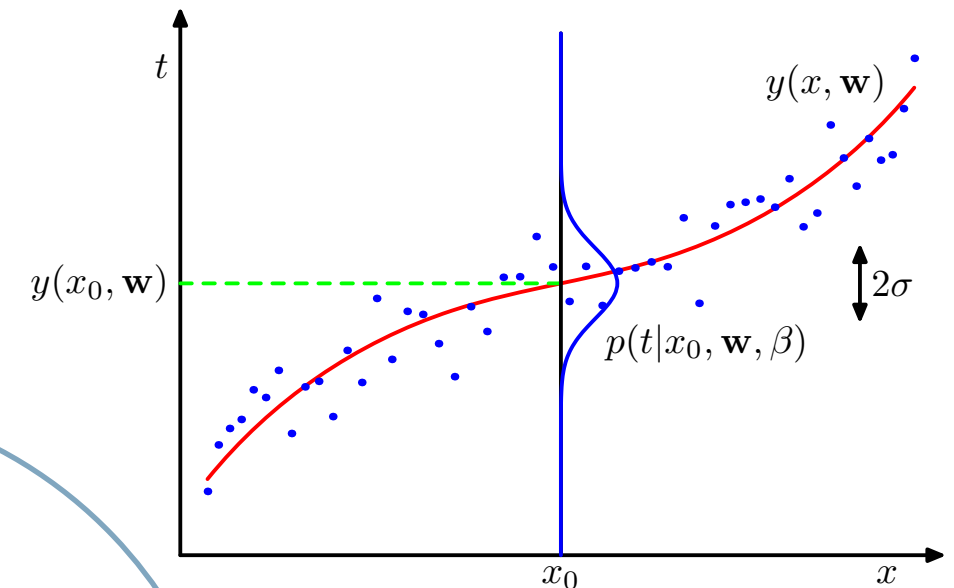
- ▶ **MAP solution** obtained by solving:

$$\mathbf{w}_{MAP} = \underset{\mathbf{w}}{\operatorname{argmin}} \quad -\log p(D | \mathbf{w}, \beta) - \log p(\mathbf{w}, \alpha)$$

MAP - Example Gaussian distribution

- ▶ **Likelihood**/Data model:

$$\begin{aligned} p(t | x, \mathbf{w}, \beta) &= \mathcal{N}(t | y(x, \mathbf{w}), \beta^{-1}) \\ &= \sqrt{\frac{\beta}{2\pi}} \exp \left[-\frac{\beta}{2} (t - y(x, \mathbf{w}))^2 \right] \end{aligned}$$



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$$\begin{aligned} \log p(D | \mathbf{w}, \beta) &= -\frac{\beta}{2} \sum_{i=1}^N (y(x_i, \mathbf{w}) - t_i)^2 \\ &\quad + \frac{N}{2} \log \beta - \frac{N}{2} \log 2\pi \end{aligned}$$

- ▶ **MAP solution** obtained by solving:

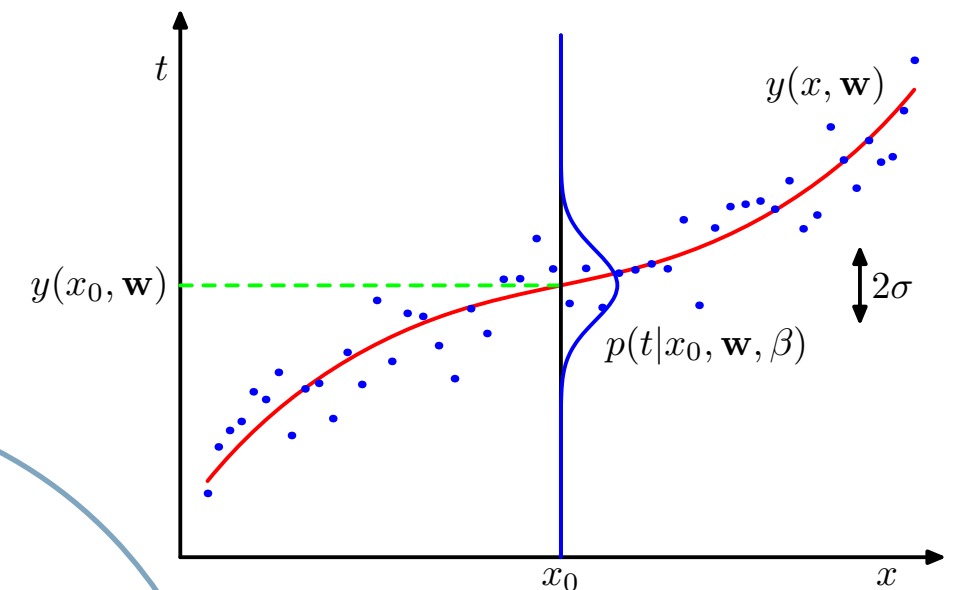
$$\mathbf{w}_{MAP} = \underset{\mathbf{w}}{\operatorname{argmin}} \frac{\beta}{2} \sum_{i=1}^N (y(x_i, \mathbf{w}) - t_i)^2 - \log p(\mathbf{w}, \alpha)$$

MAP - Example Gaussian distribution

- ▶ **Likelihood**/Data model:

$$p(t|x, \mathbf{w}, \beta) = \mathcal{N}(t|y(x, \mathbf{w}), \beta^{-1})$$

$$= \sqrt{\frac{\beta}{2\pi}} \exp \left[-\frac{\beta}{2}(t - y(x, \mathbf{w}))^2 \right]$$



- ▶ **Prior** (assumption: $\mathbf{w}$ takes on small values with certainty α):

$$p(\mathbf{w}|\alpha) = \prod_{i=1}^M \mathcal{N}(w_i|0, \alpha^{-1})$$

$$= \left(\frac{\alpha}{2\pi}\right)^{M/2} \prod_{i=1}^M e^{-\frac{\alpha}{2}w_i w_i}$$

$$= \left(\frac{\alpha}{2\pi}\right)^{M/2} e^{-\frac{\alpha}{2}\mathbf{w}^T \mathbf{w}}$$

$$\log p(D|\mathbf{w}, \beta) = -\frac{\beta}{2} \sum_{i=1}^N (y(x_i, \mathbf{w}) - t_i)^2$$

$$+ \frac{N}{2} \log \beta - \frac{N}{2} \log 2\pi$$

$$\log p(\mathbf{w}|\alpha) = -\frac{\alpha}{2} \mathbf{w}^T \mathbf{w}$$

$$+ \frac{N}{2} \log \alpha - \frac{N}{2} \log 2\pi$$

- ▶ **MAP solution** obtained by solving:

$$\mathbf{w}_{MAP} = \underset{\mathbf{w}}{\operatorname{argmin}} \frac{\beta}{2} \sum_{i=1}^N (y(x_i, \mathbf{w}) - t_i)^2 + \frac{\alpha}{2} \mathbf{w}^T \mathbf{w}$$

Parameter estimates with Gaussians

- Given **Likelihood**/Data model:

$$p(t | x, \mathbf{w}, \beta) = \mathcal{N}(t | y(x, \mathbf{w}), \beta^{-1})$$

- The **ML parameter estimate** is obtained via **least squares**:

$$\mathbf{w}_{ML} = \operatorname{argmin}_{\mathbf{w}} \frac{\beta}{2} \sum_{i=1}^N (y(x_i, \mathbf{w}) - t_i)^2$$

- Additionally, given Gaussian weight **Prior**:

$$p(\mathbf{w} | \alpha) = \prod_{i=1}^M \mathcal{N}(w_i | 0, \alpha^{-1})$$

- The **MAP parameter estimate** is obtained via **regularized least squares**:

$$\mathbf{w}_{MAP} = \operatorname{argmin}_{\mathbf{w}} \frac{\beta}{2} \sum_{i=1}^N (y(x_i, \mathbf{w}) - t_i)^2 + \frac{\alpha}{2} \mathbf{w}^T \mathbf{w}$$



Machine Learning 1

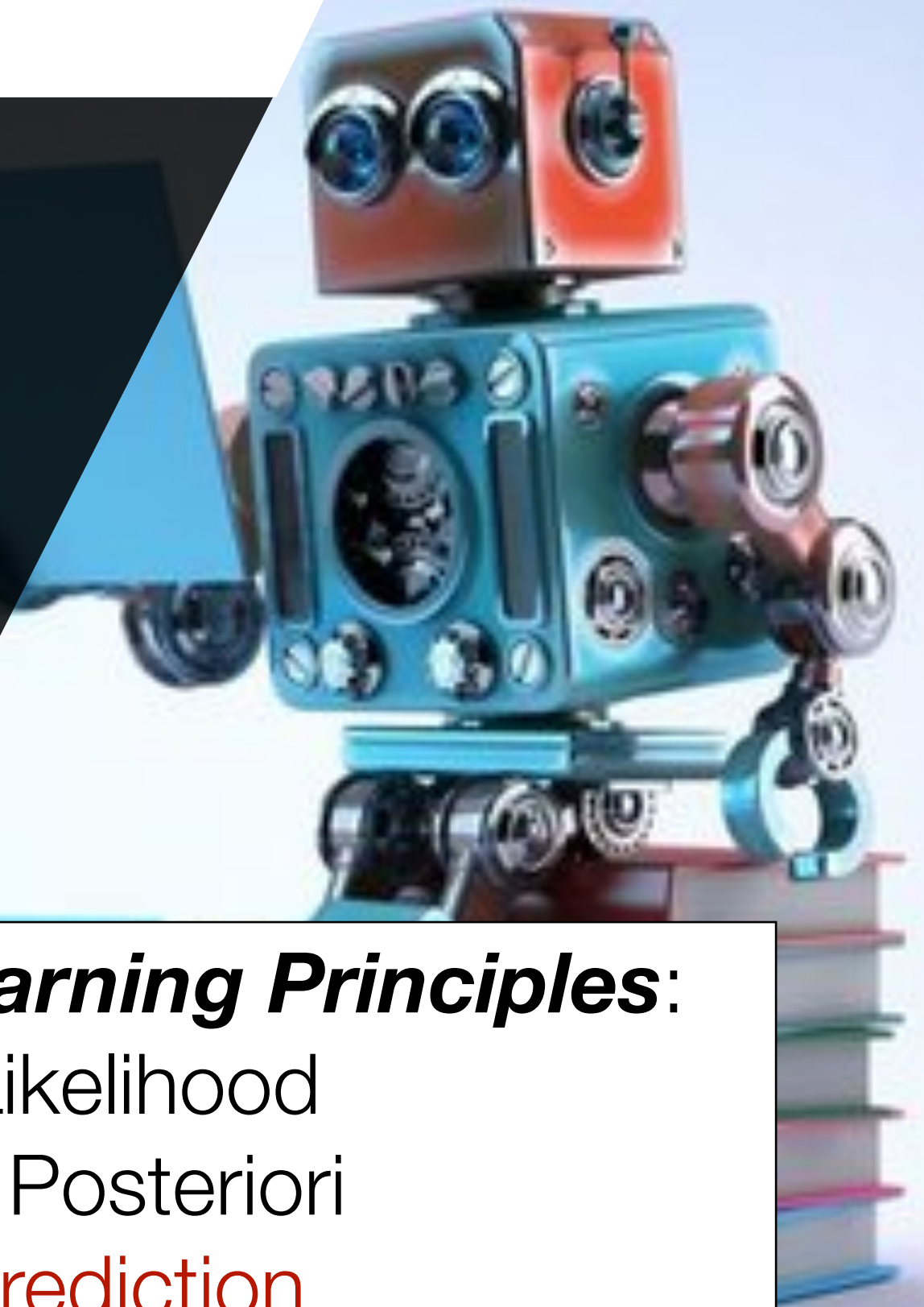
Lecture 2.6 - Bayesian Prediction

Erik Bekkers

(Bishop 1.2.6)

Three Statistical Learning Principles:

Maximum Likelihood
Maximum A Posteriori
Bayesian Prediction



Bayesian Approach

- Dataset $D = (x_1, x_2, \dots, x_N)$ of N independent observations.

- Frequentist** approach: search for **one** optimal **estimate** of $\mathbf{w}$

$$\begin{aligned}\mathbf{w}_{ML} &= \underset{\mathbf{w}}{\operatorname{argmax}} p(D | \mathbf{w}) \\ \mathbf{w}_{MAP} &= \underset{\mathbf{w}}{\operatorname{argmax}} p(\mathbf{w} | D)\end{aligned} \quad \left. \vphantom{\begin{aligned}\mathbf{w}_{ML} &= \underset{\mathbf{w}}{\operatorname{argmax}} p(D | \mathbf{w}) \\ \mathbf{w}_{MAP} &= \underset{\mathbf{w}}{\operatorname{argmax}} p(\mathbf{w} | D)\end{aligned}} \right\} \text{point estimate}$$

- Bayesian approach**: Given a prior belief over $\mathbf{w}$, $p(\mathbf{w})$, and our data D , we are interested in the entire posterior **distribution**

$$p(\mathbf{w} | D) = \frac{p(D | \mathbf{w}) p(\mathbf{w})}{p(D)}$$

Here we do not pick one value for $\mathbf{w}$ but consider all of them! We will weigh each model parameterized with $\mathbf{w}$ according to their posterior probability

- $p(\mathbf{w} | D)$ reflects the **plausibility of different $\mathbf{w}$** , given our prior knowledge and how likely our data is generated using $\mathbf{w}$.

⑤. We consider both x and $\underline{w}$ random variables according to joint prob $p(x', \underline{w} | D)$. i.e. $p(x', \underline{w} | D)$ gives the prob. of observing a new x' and model params $\underline{w}$ given the already observed dataset D .

- In the predictive setting we are only interested in x' , and thus we compute the marginal $p(x' | D)$ by integrating out $\underline{w}$.

- Following the product rule $p(x', \underline{w} | D) = p(x' | \underline{w}, D) p(\underline{w} | D)$ the marginal gets the interpretation of a weighted average sum/integral. Weighting each predictive model $p(x' | \underline{w})$ with the posterior $p(\underline{w} | D)$ for that $\underline{w}$.

Bayesian Approach

conditional independence

- ▶ Prior distribution: $p(\mathbf{w})$, should represent some prior knowledge/belief of the plausibility of $\mathbf{w}$.

- ▶ After observing data $D = (x_1, x_2, \dots, x_N)$, posterior distribution

$$p(\mathbf{w} | D) = \frac{p(D | \mathbf{w})p(\mathbf{w})}{p(D)}$$

pred dist. *posterior*

- ▶ Bayesian predictive distribution:

$$p(x' | D) = \int p(x', \mathbf{w} | D) d\mathbf{w} = \int p(x' | \mathbf{w}, D) \cdot p(\mathbf{w} | D) d\mathbf{w}$$

$$= \int p(x' | \mathbf{w}) p(\mathbf{w} | D) d\mathbf{w}$$

Curve Fitting: Bayesian Approach

- ▶ Predictive distribution:

$$p(t' | x', D) = \int p(t' | x', \mathbf{w}) p(\mathbf{w} | D) d\mathbf{w}$$

$p(D|\mathbf{w}) p(\mathbf{w}) \cdot \frac{1}{\text{const}}$

$\int_x \dots \delta_x$

Using posterior $p(\mathbf{w} | D) = \frac{p(D | \mathbf{w}) p(\mathbf{w})}{p(D)}$

with

$$p(\mathbf{t} | \mathbf{x}) = \int p(\mathbf{t} | \mathbf{x}, \mathbf{w}) p(\mathbf{w} | D) d\mathbf{w}$$

$p(\mathbf{t}, \mathbf{x})$

Advantages:

- ✓ ▶ Inclusion of prior knowledge
- ✓ ▶ Represents uncertainty in t' both due to target noise, and uncertainty over $\mathbf{w}$.

Disadvantages:

- ✓ ▶ Posterior is hard to compute analytically → approximate!
- ✓ ▶ Prior is often chosen for mathematical convenience, not reflection of prior belief!

→ Normal (Gaussian)

Conjugate priors

We say that a distribution $p(\theta)$ is a **conjugate prior** for a likelihood function $p(x | \theta)$ iff the posterior distribution $p(\theta | x)$ is in the same probability distribution family as the prior

Examples of conjugate distributions:

Likelihood	Conjugate prior
Gaussian	Gaussian
Bernoulli	Beta
Poisson	Gamma
Uniform	Pareto

...

Three Statistical Learning Principles

Three general statistical learning principles to go from data to models (parametric predictive/proposal distributions):

- I. Maximum likelihood
- II. Maximum a posteriori
- III. Bayesian prediction



Machine Learning 1

Lecture 3.1 - Supervised Learning

Linear Regression With Basis Functions

Erik Bekkers

(Bishop 3.1)





Machine Learning 1

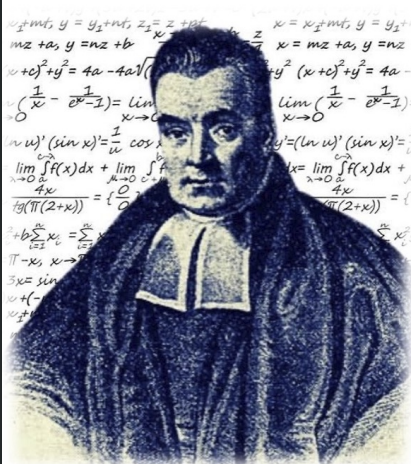
Lecture 3.1 - Supervised Learning

Linear Regression With Basis Functions

Erik Bekkers

(Bishop 3.1)

The probabilistic view



W7

Gaussian Processes

W6

Kernel Methods

W5

Latent variable models

W4

Neural Networks

W3

Classification

W2

Regression

W1

Probabilistic modeling

Random variables, distributions,
maximum likelihood estimation,
MAP

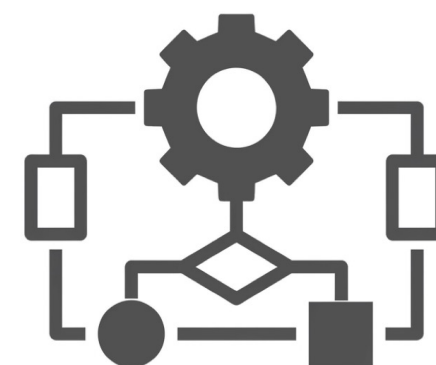
Optimization

Exact/analytic, (stochastic
gradient descent, constrained
optimization, 2nd order)

Mathematical tools

Multi-variate calculus, linear
algebra

The algorithmic view



Linear Regression

▶ Regression on dataset $D = \{(\mathbf{x}_1, t_1), \dots, (\mathbf{x}_N, t_N)\}$

▶ Input variables $\mathbf{x}_n \in \mathbb{R}^d$

▶ Target variables $t_n \in \mathbb{R}$

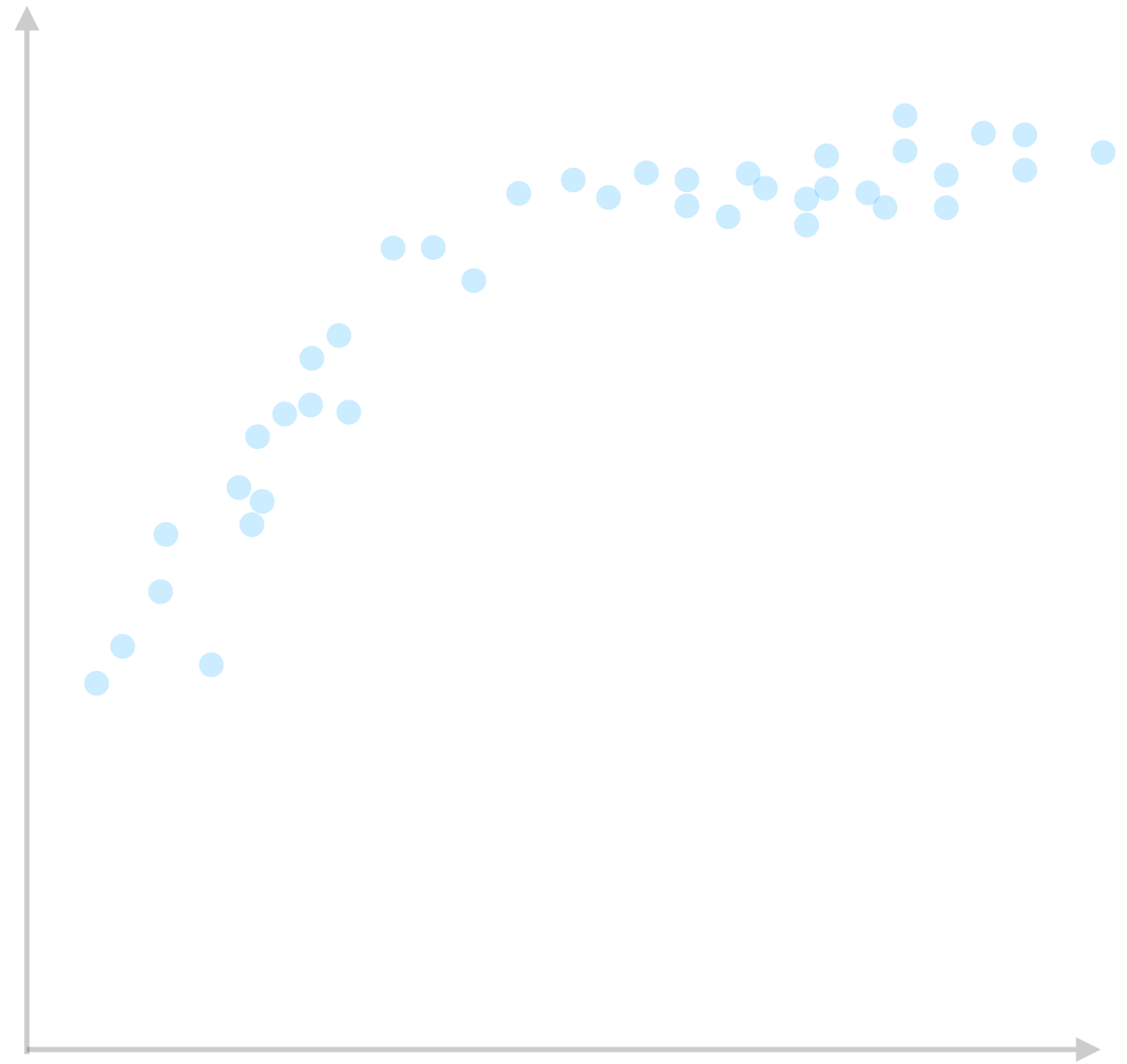
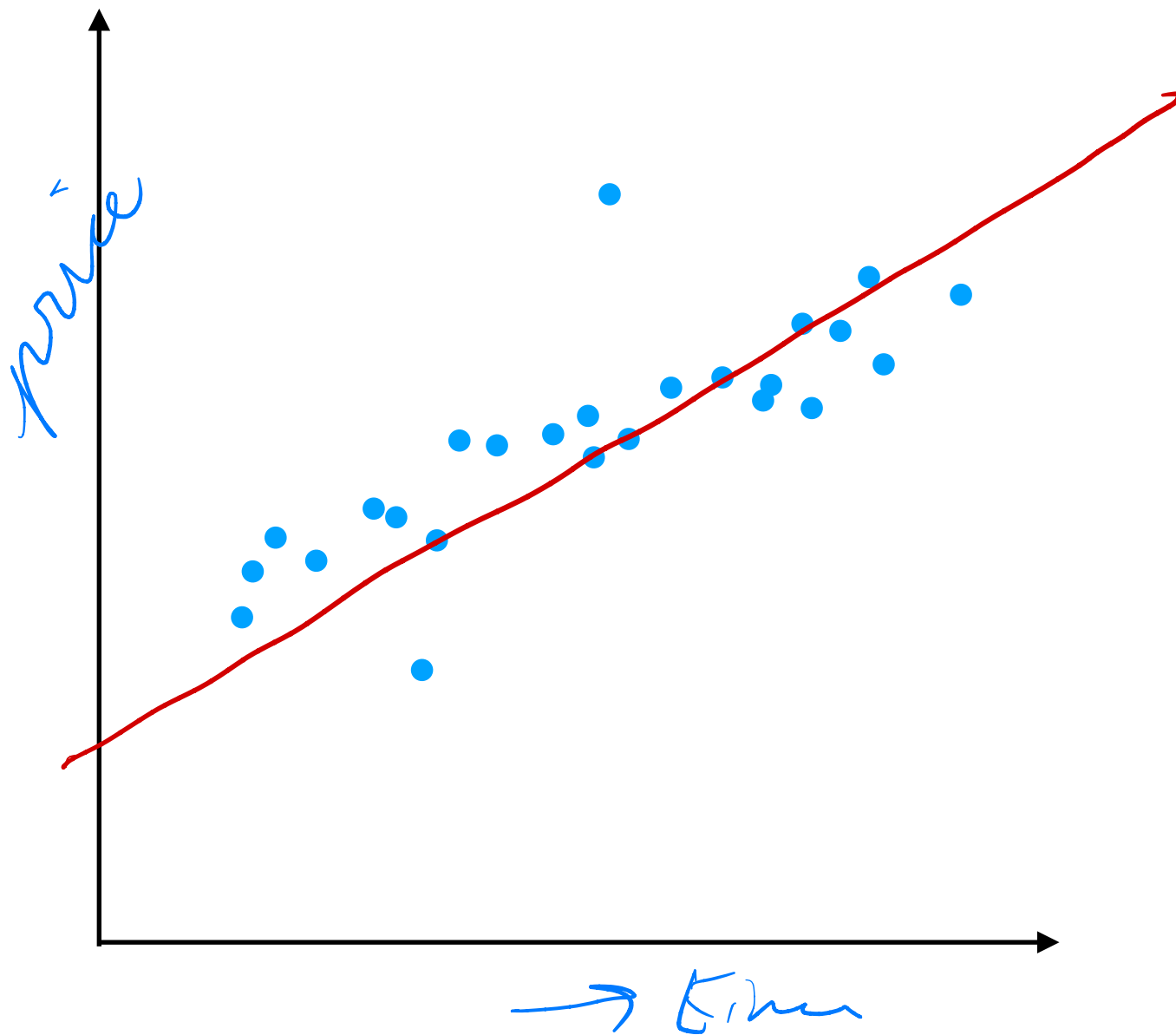
$$\underline{X} = \begin{pmatrix} x_{n1} \\ x_{n2} \\ x_{n3} \\ \vdots \\ x_{nd} \end{pmatrix} \in \mathbb{R}^d, \quad \underline{w} = \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_d \end{pmatrix}$$

▶ Simplest linear model:

$$y(\mathbf{x}, \mathbf{w}) = w_0 + w_1 x_1 + w_2 x_2 + \dots + w_D x_D$$

(bias)

Linear Regression



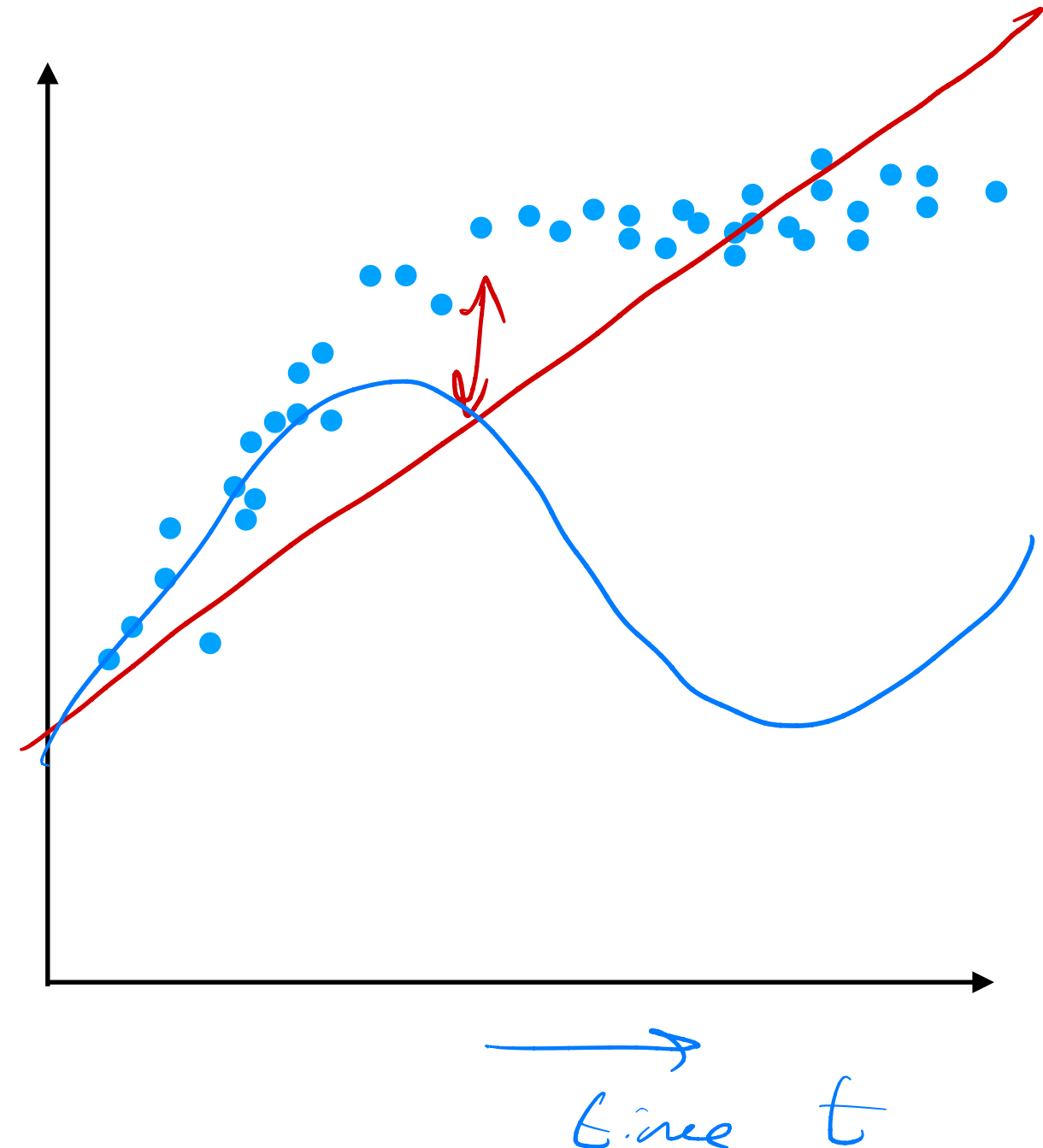
Linear Regression

$$\phi_1(t) = t$$

$$\phi_2(t) = t^2$$

$$\phi_3(t) = \sin(t)$$

$$y(t) = w_1 \phi_1(t) + w_2 \phi_2(t) + w_3 \phi_3(t)$$



Linear Basis Models

- Fix number of parameters M s.t.

$$\mathbf{w} = \begin{pmatrix} w_0 \\ w_1 \\ \vdots \\ w_{M-1} \end{pmatrix} \in \mathbb{R}^M$$

$\mathbb{R}^{d=1}$

- Choose $M - 1$ basis functions/features of $\mathbf{x}$:

$$\phi_i(\underline{x}) \in \mathbb{R}$$

$$\phi_i: \mathbb{R}^d \rightarrow \mathbb{R}$$

- Model:

$$y(\mathbf{x}, \mathbf{w}) = w_0 + \sum_{m=1}^{M-1} w_m \phi_m(\underline{x})$$

w_0 : bias

- Set $\phi_0(\mathbf{x}) = 1$ such that $\boldsymbol{\phi}(\mathbf{x}) = (1, \phi_1(x), \phi_2(x), \dots, \phi_{M-1}(x))^T$

$$y(\mathbf{x}, \mathbf{w}) = \underline{\mathbf{w}}^T \underline{\boldsymbol{\phi}}(\underline{x})$$

Example: Basis Functions (I)

- ▶ **Standard basis**/projection on input components:
for $\mathbf{x} = (x_1, x_2, \dots, x_D)^T$:

$$\phi_i(\mathbf{x}) = x_i$$

$$y(\mathbf{x}, \mathbf{w}) = w_0 + \sum_{i=1}^M w_i \phi_i(\mathbf{x}) = w_0 + w_1 x_1 + w_2 x_2 + \dots$$

- ▶ **Polynomial basis**/i-power map for $x \in \mathbb{R}$:

$$\phi_i(x) = x^i$$

$$y(x, \mathbf{w}) = w_0 + w_1 \cdot x + w_2 \cdot x^2 + w_3 \cdot x^3 + \dots$$

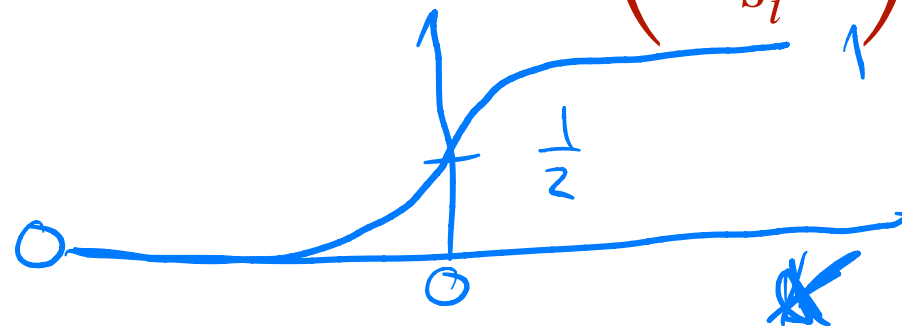
Example: Basis Functions (II)

- ▶ Gaussian basis functions with $\mathbf{x} \in \mathbb{R}^D$: $\phi_i(\mathbf{x}) = \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}_i^{-1}(\mathbf{x} - \boldsymbol{\mu}_i)\right)$


$$y(\mathbf{x}, \mathbf{w}) = w_0 + \sum_{i=1}^{M-1} w_i \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}_i^{-1}(\mathbf{x} - \boldsymbol{\mu}_i)\right)$$

- ▶ Logistic sigmoid functions with $x \in \mathbb{R}$: $\phi_i(x) = \sigma\left(\frac{x - \mu_i}{s_i}\right)$

with $\sigma(x) = \frac{1}{1 + e^{-x}}$



$$y(\mathbf{x}, \mathbf{w}) = w_0 + \sum_{i=1}^{M-1} w_i \sigma\left(\frac{x - \mu_i}{s_i}\right)$$

Example: Basis Functions (III)

<https://ideal-machine-intelligence.eu/teaching/ml1/explainers/maximum-likelihood.html>



Machine Learning 1

Lecture 3.2 - Supervised Learning

Linear Regression via Maximum Likelihood
Optimization

Erik Bekkers

(Bishop 3.1.1)



Linear Regression

- ▶ Regression $D = \{(\mathbf{x}_1, t_1), \dots, (\mathbf{x}_N, t_N)\}$
- ▶ Input variables $\mathbf{x}_i \in \mathbb{R}^D$
- ▶ Target variables t_i

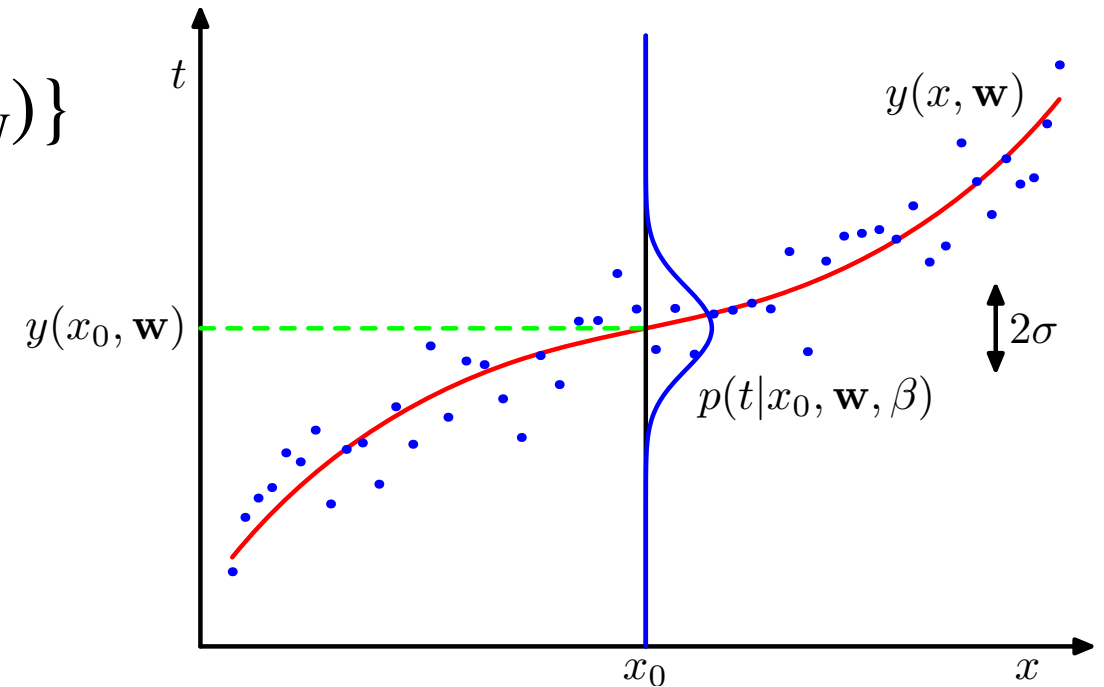


Figure: Gaussian conditional distribution (Bishop 1.16)

- ▶ Linear model with basis functions

$$y(\mathbf{x}, \mathbf{w}) = \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x})$$

$\mathbb{R}^m$

Maximum Likelihood

► **Model assumptions/choices:**

- Data is generated through model $y(\mathbf{x}, \mathbf{w}) = \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x})$
- Observations t_i are subject to Gaussian noise

$$t = y(\mathbf{x}, \mathbf{w}) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \beta^{-1})$$

► Then **data distribution (likelihood):**

$$p(t | \mathbf{x}, \mathbf{w}, \beta) = \mathcal{N}(t | y(\mathbf{x}, \mathbf{w}), \beta^{-1})$$

$$\begin{pmatrix} -x_1- \\ -x_i- \end{pmatrix} \in \mathbb{R}^{N \times D}$$

► **Dataset:** $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_N)^T$ and $\mathbf{t} = (t_1, \dots, t_N)^T \leftarrow \mathbb{R}^N$

► **Likelihood function:**

$$p(\mathbf{t} | \mathbf{X}, \mathbf{w}, \beta) = \prod_{i=1}^N \sqrt{\frac{\beta}{2\pi}} e^{-\frac{\beta}{2}(t_i - \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_i))^2}$$

ML: Sum-of-Squares Error

- ▶ Likelihood:

$$p(\mathbf{t} | \mathbf{X}, \mathbf{w}, \beta) = \prod_{i=1}^N \mathcal{N}(t_i | \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_i), \beta^{-1})$$

- ▶ Log likelihood

$$\log p(\mathbf{t} | \mathbf{X}, \mathbf{w}, \beta) = \frac{N}{2} \log \beta - \frac{N}{2} \log 2\pi - \frac{\beta}{2} \sum_{i=1}^N (t_i - \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_i))^2$$

- ▶ Sum-of-squares error:

$$E_D(\mathbf{w}) = \frac{1}{2} \sum_{i=1}^N (t_i - \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_i))^2$$

- ▶ For comparison of different dataset sizes N

$$E_D^{RMSE}(\mathbf{w}) = \sqrt{\frac{1}{N} \sum_{i=1}^N (t_i - \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_i))^2}$$

+ no dependency on dataset size N
+ sensible error in units, e.g. euros instead of squared euros

Example: Sum-of-Squares Error

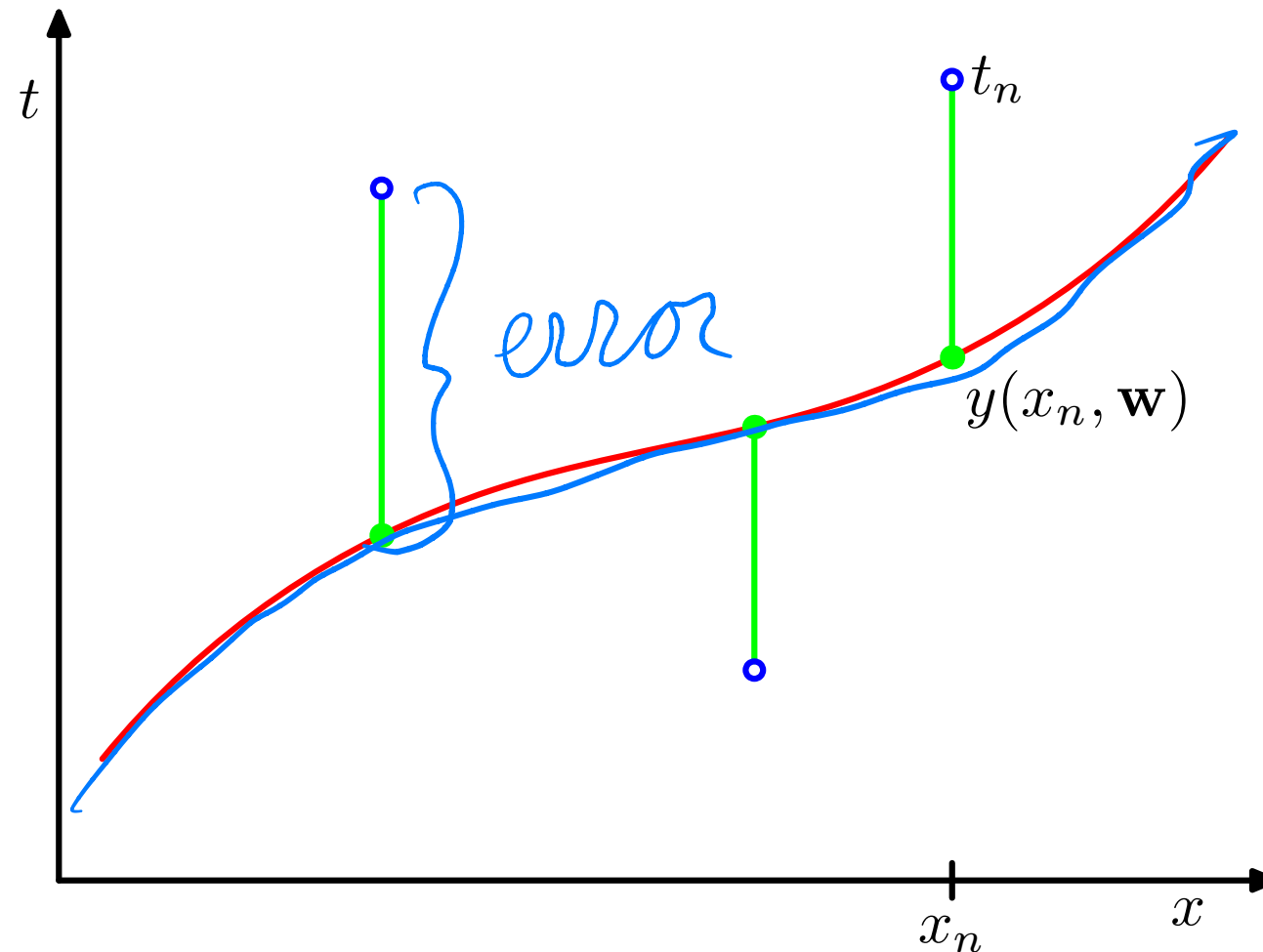


Figure: Errors are given by half the squares of green bars (Bishop 1.3)

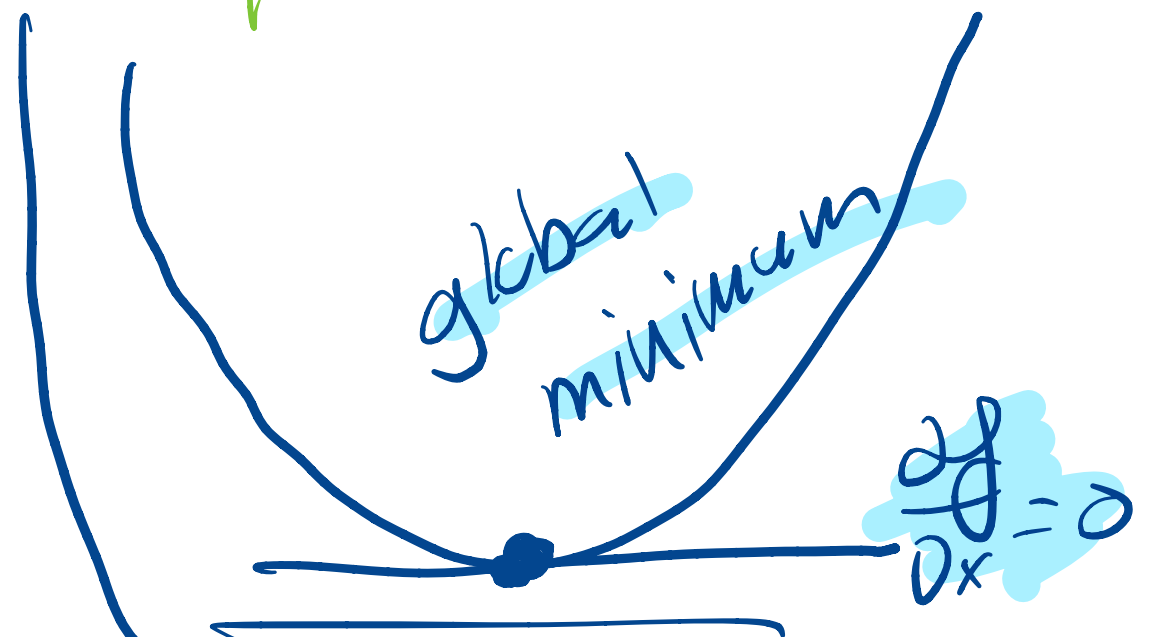
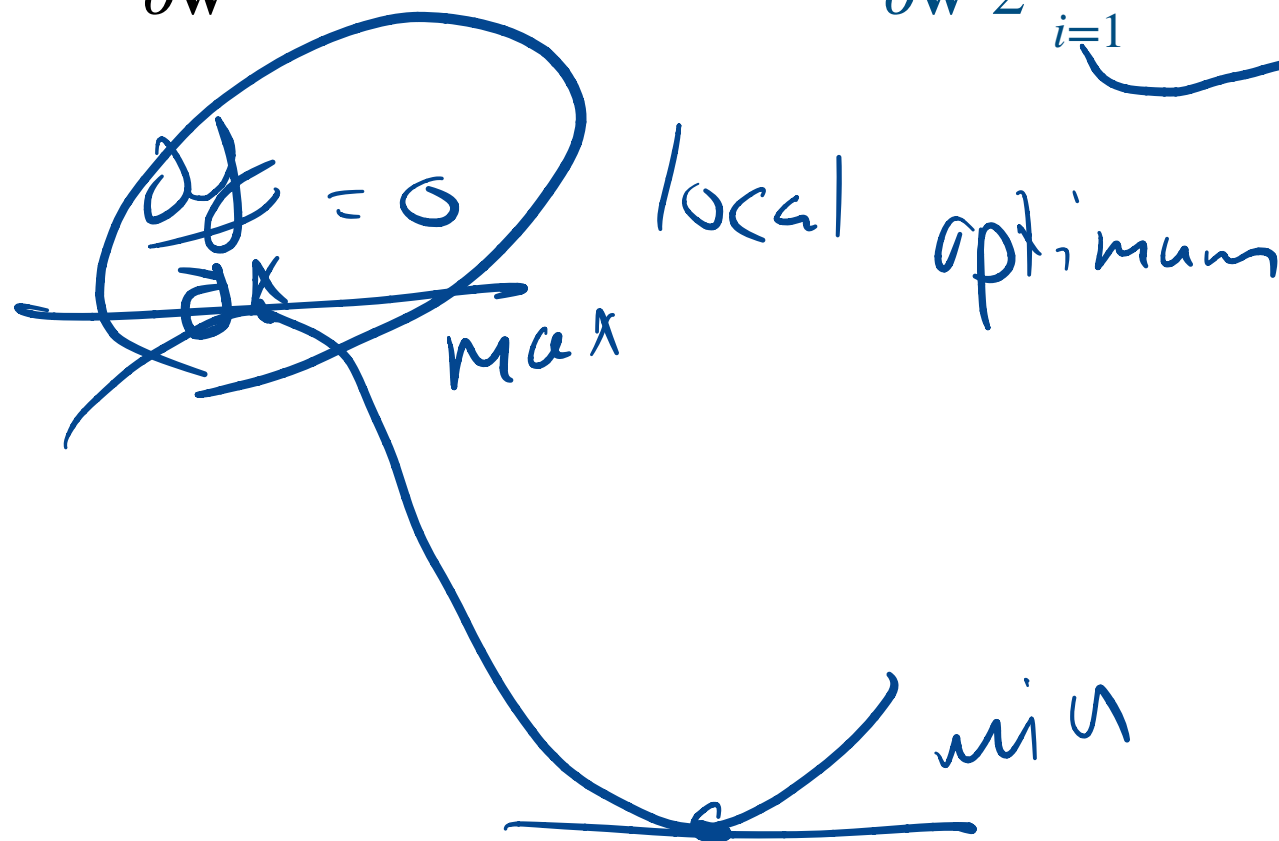
Maximum Likelihood Estimates

- Maximize the log likelihood $\log p(\mathbf{t} | \mathbf{X}, \mathbf{w}, \beta)$

- Analytic solution by solving: $\frac{\partial}{\partial \mathbf{w}} \log p(\mathbf{t} | \mathbf{X}, \mathbf{w}, \beta) = 0$ for $\mathbf{w} \in \mathbb{R}^M$

$$\frac{\partial}{\partial \mathbf{w}} \log p(\mathbf{t} | \mathbf{X}, \mathbf{w}, \beta) = -\beta \frac{\partial}{\partial \mathbf{w}} \frac{1}{2} \sum_{i=1}^N \{t_i - \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_i)\}^2 = \mathbf{0}^T$$

quadratic in $\mathbf{w} \Rightarrow$ convex



since convex in $\mathbf{w}$

Convention $\nabla_{\mathbf{x}} f(\mathbf{x}) := \frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} := \left(\frac{\partial f(\mathbf{x})}{\partial x_1}, \frac{\partial f(\mathbf{x})}{\partial x_2}, \dots \right)$ is a row vector

Maximum Likelihood Estimates

- Maximize the log likelihood $\log p(\mathbf{t} | \mathbf{X}, \mathbf{w}, \beta)$

- Analytic solution by solving: $\frac{\partial}{\partial \mathbf{w}} \log p(\mathbf{t} | \mathbf{X}, \mathbf{w}, \beta) = 0$ for $\mathbf{w}$

$$\frac{\partial}{\partial \mathbf{w}} \log p(\mathbf{t} | \mathbf{X}, \mathbf{w}, \beta) = -\beta \frac{\partial}{\partial \mathbf{w}} \frac{1}{2} \sum_{i=1}^N \{t_i - \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_i)\}^2 = \mathbf{0}^T$$

$(\beta > 0, \text{ and apply derivative})$
 $\Leftrightarrow \sum_{i=1}^N \{t_i - \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_i)\} \boldsymbol{\phi}(\mathbf{x}_i)^T = \mathbf{0}^T$

$$\underline{a}^T \underline{b} = \underline{b}^T \underline{a}$$

$$\frac{\partial}{\partial \underline{w}} (\underline{w}^T \boldsymbol{\phi})$$

$$\frac{\partial}{\partial \underline{w}} (\boldsymbol{\phi}^T \underline{w}) = \boldsymbol{\phi}^T$$

Convention $\nabla_{\mathbf{x}} f(\mathbf{x}) := \frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} := \left(\frac{\partial f(\mathbf{x})}{\partial x_1}, \frac{\partial f(\mathbf{x})}{\partial x_2}, \dots \right)$ is a row vector

Maximum Likelihood Estimates

- Maximize the log likelihood $\log p(\mathbf{t} | \mathbf{X}, \mathbf{w}, \beta)$
- Analytic solution by solving: $\frac{\partial}{\partial \mathbf{w}} \log p(\mathbf{t} | \mathbf{X}, \mathbf{w}, \beta) = \mathbf{0}$ for $\mathbf{w}$

$$\frac{\partial}{\partial \mathbf{w}} \log p(\mathbf{t} | \mathbf{X}, \mathbf{w}, \beta) = -\beta \frac{\partial}{\partial \mathbf{w}} \frac{1}{2} \sum_{i=1}^N \{t_i - \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_i)\}^2 = \mathbf{0}^T$$

$(\beta > 0, \text{ and apply derivative})$
 $\Leftrightarrow \sum_{i=1}^N \{t_i - \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_i)\} \boldsymbol{\phi}(\mathbf{x}_i)^T = \mathbf{0}^T$

(rearrange terms)
 $\Leftrightarrow \sum_{i=1}^N \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_i) \boldsymbol{\phi}(\mathbf{x}_i)^T = \sum_{i=1}^N t_i \boldsymbol{\phi}(\mathbf{x}_i)^T$

Convention $\nabla_{\mathbf{x}} f(\mathbf{x}) := \frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} := \left(\frac{\partial f(\mathbf{x})}{\partial x_1}, \frac{\partial f(\mathbf{x})}{\partial x_2}, \dots \right)$ is a row vector

Maximum Likelihood Estimates

- Maximize the log likelihood $\log p(\mathbf{t} | \mathbf{X}, \mathbf{w}, \beta)$

- Analytic solution by solving: $\frac{\partial}{\partial \mathbf{w}} \log p(\mathbf{t} | \mathbf{X}, \mathbf{w}, \beta) = 0$ for $\mathbf{w}$

$$\frac{\partial}{\partial \mathbf{w}} \log p(\mathbf{t} | \mathbf{X}, \mathbf{w}, \beta) = -\beta \frac{\partial}{\partial \mathbf{w}} \frac{1}{2} \sum_{i=1}^N \{t_i - \mathbf{w}^T \phi(\mathbf{x}_i)\}^2 = \mathbf{0}^T$$

($\beta > 0$, and apply derivative)

$$\sum_{i=1}^N \{t_i - \mathbf{w}^T \phi(\mathbf{x}_i)\} \phi(\mathbf{x}_i)^T = \mathbf{0}^T$$

(rearrange terms)

$$\sum_{i=1}^N \mathbf{w}^T \phi(\mathbf{x}_i) \phi(\mathbf{x}_i)^T = \sum_{i=1}^N t_i \phi(\mathbf{x}_i)^T$$

(transpose both sides)

$$\left(\sum_{i=1}^N \phi(\mathbf{x}_i) \phi(\mathbf{x}_i)^T \right) \mathbf{w} = \sum_{i=1}^N t_i \phi(\mathbf{x}_i)$$

$M \times M$ $M \times 1$

Maximum Likelihood Estimates

- So, optimal $\mathbf{w}_{ML}$ satisfies

$$\left(\sum_{i=1}^N \phi(\mathbf{x}_i) \phi(\mathbf{x}_i)^T \right) \mathbf{w} = \sum_{i=1}^N \phi(\mathbf{x}_i) t_i$$

- Introduce the design matrix Φ

$$\Phi \in \mathbb{R}^{N \times M}$$

$$\Phi = (\phi(\mathbf{x}_1), \phi(\mathbf{x}_2), \dots)^T = \begin{pmatrix} \phi_0(\mathbf{x}_1) & \phi_1(\mathbf{x}_1) & \dots & \phi_{M-1}(\mathbf{x}_1) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(\mathbf{x}_N) & \phi_1(\mathbf{x}_N) & \dots & \phi_{M-1}(\mathbf{x}_N) \end{pmatrix}^T$$

- Solve linear system

$$(\Phi^T \Phi) \mathbf{w} = \Phi^T \mathbf{t}$$

Pseudo-inverse
Moore-Penrose

$$\Rightarrow \mathbf{w}_{ML} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{t}$$

- $\mathbb{E}[t' | \mathbf{x}', \mathbf{w}_{ML}] = y(\mathbf{x}', \mathbf{w}_{ML}) =$

$$\mathbf{w}_{ML}^T \phi(\mathbf{x}')$$



Machine Learning 1

Lecture 3.3 - Supervised Learning
Stochastic Gradient Descent

Erik Bekkers

(Bishop 3.1.3)





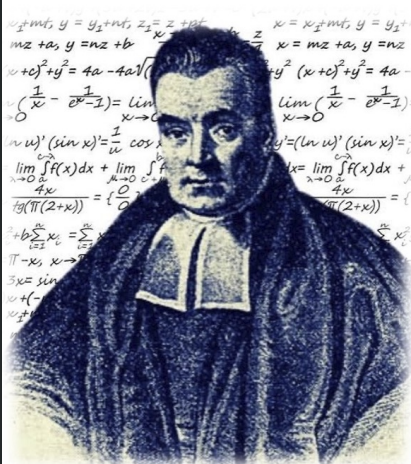
Machine Learning 1

Lecture 3.3 - Supervised Learning Stochastic Gradient Descent

Erik Bekkers

(Bishop 3.1.3)

The probabilistic view



W7

Gaussian Processes

W6

Kernel Methods

W5

Latent variable models

W4

Neural Networks

W3

Classification

W2

Regression

W1

Probabilistic modeling

Random variables, distributions,
maximum likelihood estimation,
MAP

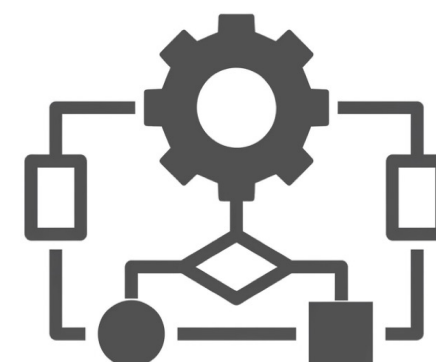
Optimization

Exact/analytic, (stochastic
gradient descent, constrained
optimization, 2nd order)

Mathematical tools

Multi-variate calculus, linear
algebra

The algorithmic view



Stochastic gradient descent

- For $N \gg 1$ solution $\mathbf{w}_{ML} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{t}$ is very costly to compute!

- Needs to process all data $(\mathbf{x}_1, \dots, \mathbf{x}_N)$ at once.

- Matrix inversion of $M \times M$ matrix: $O(M^3)$

Expensive

- Loss is a sum of error terms for each datapoint:

- $$E_D(\mathbf{w}) = \sum_{i=1}^N E(\mathbf{x}_i, t_i, \mathbf{w})$$

- $$E(\mathbf{x}_i, t_i, \mathbf{w}) = \frac{1}{2} (t_i - \mathbf{w}^T \phi(\mathbf{x}_i))^2$$

- Approach for large dataset: stochastic gradient descent

idea approximate E_D with few data points

Recap: The Gradient (see e.g. Bishop App. E)

- ▶ The gradient encodes all directional derivatives via scalar product

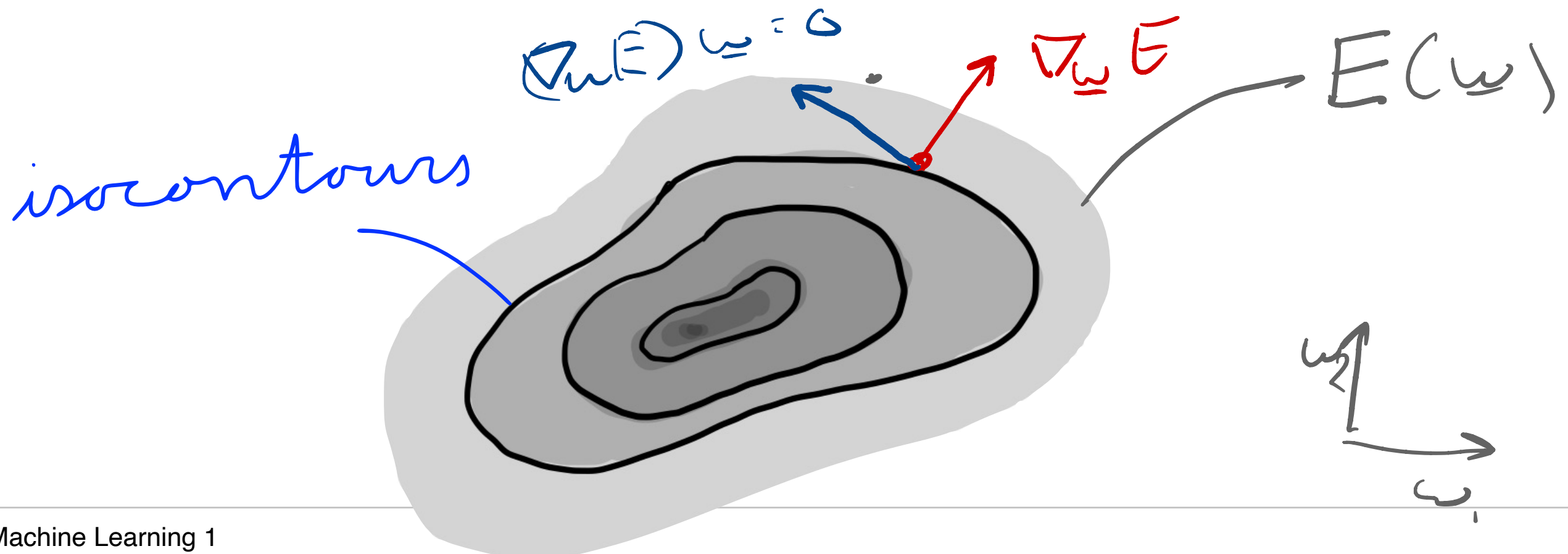
$$\nabla_{\underline{w}} E := \left(\frac{\partial E}{\partial w_1}, \frac{\partial E}{\partial w_2}, \dots \right) \quad \text{a row vector}$$

directional derivative $\underline{v}$ in direction $\underline{v}$ is given by $(\nabla_{\underline{w}} E) \underline{v}$

- ▶ The gradient always points in the direction of steepest ascent

directional derivative is maximized when $\underline{v} = \nabla_{\underline{w}} E$

- ▶ The gradient is always perpendicular to the contours of a function



Stochastic gradient descent

- Objective: minimize $E_D(\mathbf{w}) = \sum_{i=1}^N E(\mathbf{x}_i, t_i, \mathbf{w})$, with $E(\mathbf{x}_i, t_i, \mathbf{w}) = \frac{1}{2}(t_i - \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_i))^2$

- ▶ Stochastic gradient descent (SGD):
 - ▶ Initialize $\mathbf{w}^{(0)}$, choose learning rate η
 - ▶ Iterate over data points, and update

$$\begin{aligned}\mathbf{w}^{(\tau+1)} &= \mathbf{w}^{(\tau)} - \eta(\nabla_{\mathbf{w}} E(\mathbf{x}_i, t_i, \mathbf{w}^{(\tau)}))^T \\ &= \mathbf{w}^{(\tau)} - \eta(t_i - \phi(\mathbf{x}_i)^T \mathbf{w}^{(\tau)})\phi(\mathbf{x}_i)\end{aligned}$$

Often you pick a (mini-) batch of samples B and compute the gradient on this batch instead of a single sample.

- ▶ If $E_D(\mathbf{w})$ is convex in $\mathbf{w}$ and η is small enough: **convergence**

